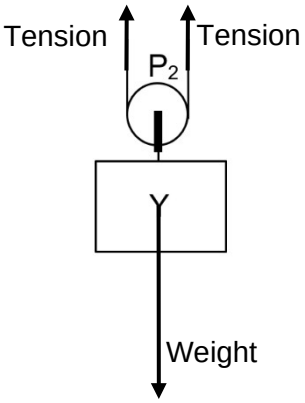
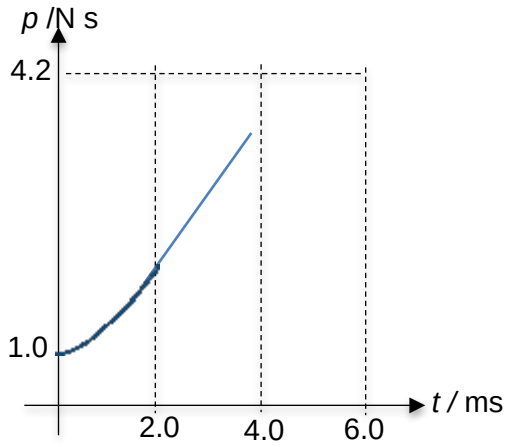
Tampines Meridian Junior College

2026 JC2 H2 Physics Preliminary Examination Paper 3

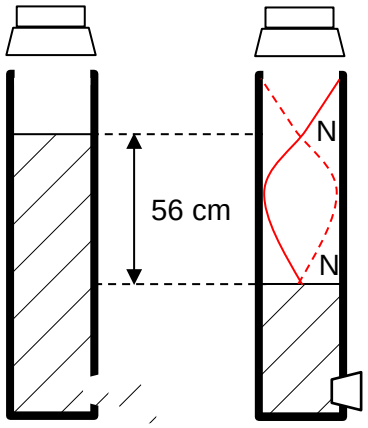
Suggested Solution

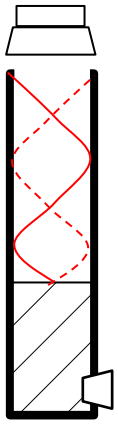
1	**(a)**	For LHS: $[F] = \text{kg m s}^{-2}$ [M1] For RHS: $[\rho A v^2] = \text{kg m}^{-3}\ (\text{m}^2)(\text{m s}^{-1})^2 = \text{kg m s}^{-2}$ [M1] Since unit of LHS = units of RHS, hence equation is homogenous. [A1] Can accept: [N] = [kg] [m s^{-2}]
	(b)	[B2: any two of the following possible random errors] There might be possible fluctuations in speed, air condition (density of air) and even cross-sectional area of the cyclist when riding. There could also be variation to the start or/and stop time of stopwatch.
	(c)	A zero error in the force sensors affects the accuracy of the measurements, but not the precision. [B1] This can be corrected by subtracting/adding the zero error from all measured values, or by recalibrating the sensor before taking measurements. [B1]

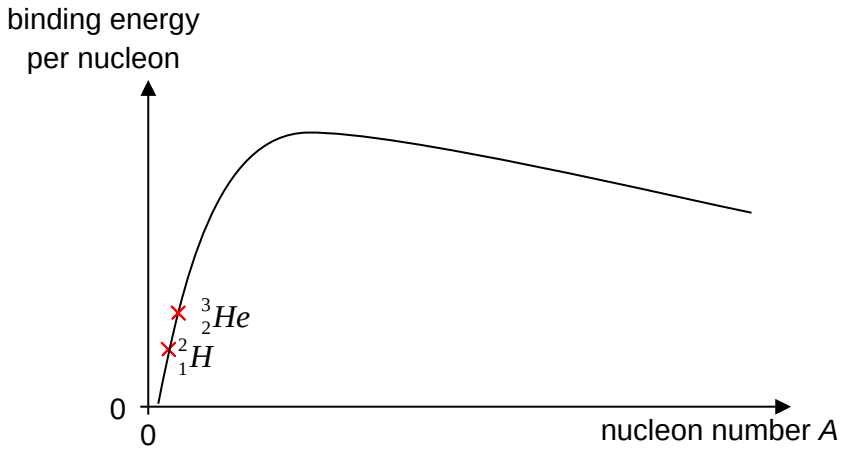
2	**(a)**	Newton's first law of motion states that a body at rest will stay at rest, and a body in motion will continue to move at constant velocity, unless acted on by a resultant external force.		A1
	(b)	**(i)**	work done by $f = 5.0 \times 1.5 \cos 180^\circ = -7.5$ J	A1
		(ii)	By conservation of energy, total kinetic energy = $m_x g\Delta h_x + m_y g\Delta h_y$ + work done by friction $= (2.0)(9.81)(1.5 \cos 60^\circ) + (4.0)(9.81)(0.75) - 7.5$ $= 36.6$ J ≈ 37 J (allow ecf for work done by friction) (allow use of kinematics equations to find velocities and total kinetic energy)	C1 A1
		(iii)	 Tensions and Weight drawn in the correct directions and positions and both labelled. (Do not accept weight labelled as *W*.) (Ignore magnitude of the forces.)	A1
		(iii)	$T + m_x g \cos 60^\circ - f = m_x a_x$ $T + (2.0)g \cos 60^\circ - 5.0 = (2.0)(2a_y)$ --- eqn 1 $m_y g - T - T = m_y a_y$ $4.0g - 2T = (4.0)a_y$ --- eqn 2 Solving, $T = 11.477$ N ≈ 11.5 N or 11 N	M1 M1 A1

3	**(a)**	The <u>total momentum of a system</u> of objects <u>remains constant</u> provided <u>no resultant external force</u> acts on the system. [B1]		
	(b)	Impulse is given by the area under the force-time graph (for a body).		[B1]
	(b)	**(i)**	Area under the *F-t* graph for 6.0 s = 0.5(0.002+0.006)(800) = 3.2 Ns	[C1]
			$\Delta p = p_{final} - p_{inital}$ $p_{final} = \Delta p + mv_{inital}$ $= 3.2 + 0.20(5.0)$ [C1] $= 4.2$ Ns [A1]	
		(ii)	 **Fig. 3.2** 1m for correct shape showing that for the first 2.0 s, (zero gradient initially and then) gradient increase/steeper, for the next 2.0 s, straight sloped line with no kinks, for the last 2.0 s gradient decrease/getting gentler (ending with zero gradient). Graph MUST be smooth with no kinks. 1m for labelling of initial *p* and final *p* value	
	(d)	There is a net <u>external force</u> that is acting <u>on the ball by the wall</u> during collision. [M1] Hence by the Principle of Conservation of Momentum, the <u>momentum of the ball is not conserved</u>. [A1]		

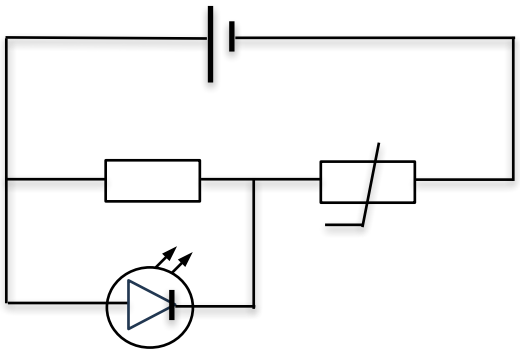
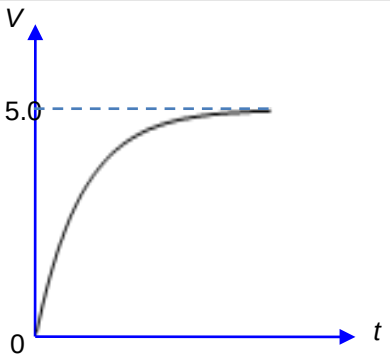
4	**(a)**	Gravitational field strength at a point is defined as the <u>gravitational force</u> <u>per unit mass</u> acting on a <u>small test mass</u> placed at that point.		A1 A1
	(b)	Gravitational field strength at a point is the <u>negative potential gradient</u> at that point.		A1
	(c)	**(i)**	gravitational field strength $g_P = \frac{GM_E}{r^2} = \frac{(6.67 \times 10^{-11})(5.98 \times 10^{24})}{(6.7 \times 10^6)^2}$ $= 8.89$ N kg^{-1}	C1 A1
	(c)	**(ii)**	$F = mg = 2200 \times 8.89 = 19600$ N	A1
		(iii)	rate of change of potential energy with distance K_P $= \lvert dU/dr \rvert = F = m\,g = 2200 \times 8.89$ $= 19600$ J m^{-1}	A1
		(iv)	Total KE $= \frac{1}{2}mv^2 = \frac{1}{2}m(GM\,r^{-1})$ -- or derive E_T = Total KE + Total GPE $= \frac{GMm}{2r} + \left(-\frac{GMm}{r}\right)$ $= -\frac{GMm}{2r}$ $= -\frac{(6.67 \times 10^{-11})(5.98 \times 10^{24})(2200)}{2(6.7 \times 10^6)}$ $= -6.6 \times 10^{10}$ J	C1 C1 A1

5	(a)	(i)	The dotted curve is the variation with diffraction of intensity for the single-slit diffraction pattern.	[A1]
		(ii)	$b\theta \approx \lambda$ $\lambda = (1.0 \times 10^{-6})(0.45)$ $= 450 \times 10^{-9}$ m *note, due to the gentle curve on the graph, and the usage of equation $b\sin\theta = \lambda$ which gives a slightly different result, wavelength ranges of 450 – 500 nm were accepted with 480nm being the median.	[M1] [A0]
		(iii)	Using n = 2 (second order maxima) (ecf λ from part (b)) $d\sin\theta = n\lambda$ $d = \frac{2(450 \times 10^{-9})}{\sin(0.25)}$ $= 3.6 \times 10^{-6}$ m Or fringe separation $x = \frac{3.2 \times 0.25}{2} = 0.4$ m $a = \frac{\lambda D}{x}$ $= 3.6 \times 10^{-6}$ m Accept a range from 3.47×10^{-6} to 3.80×10^{-6} for the value of a If n=1 is used, award 1 mark only.	[C1] [A1] [C1] [A1]
		(iv)	1. The intensity of the maxima increases or bright fringe is brighter (due to more slits). 2. Each of the maxima is narrower or fringe becomes sharper.	[B1] [B1]
	(b)	(i)		

			[A1] for shape [A1] for two nodes	
		(ii)	56 cm is equivalent to half a wavelength, thus $\lambda = 0.56 \times 2 = 1.12$ m $v = f\lambda = 310 \times 1.12$ $= 347$ m s^{-1}	[C1] [A1]
		(iii)	When the frequency is increased, the next standing wave formed looks like this:  The length from the water level to the end of the tube is $56 + \frac{56}{2} = 84$ cm This length is equivalent to $\frac{5}{4}\lambda$ Thus, $\lambda = \frac{4}{5} \times 0.84 = 0.672$ m $f = \frac{v}{\lambda} = \frac{347}{0.672} = 516$ Hz	 [C1] [M1] [A1]

6	**(a)**		Mass defect is the <u>difference in mass between</u> the mass of the individual / <u>separated protons and neutrons and the (combined) nucleus.</u>	[A1]
	(b)	**(i)**	There are 2 protons and 1 neutron in the nuclide helium−3. $\text{mass defect} = 2 \times 1.00728u + 1.00867u - 3.014931u$ $= 0.008299u$ $\text{binding energy} = \dfrac{0.008299 \times 1.66 \times 10^{-27} \times (3 \times 10^{8})^{2}}{1.6 \times 10^{-19}}$ $= 7749191\ \text{eV}\ (\text{or } 1.239 \times 10^{-12}\ \text{J})$ $\text{binding energy per nucleon} = \dfrac{7749191\ \text{eV}}{3 \times 1 \times 10^{6}}$ $= 2.58\ \text{MeV}$ $= 2.6\ \text{MeV}$ B1 for finding the mass defect B1 for finding the binding energy in J or eV M1 for finding the binding energy per nucleon in MeV	[B1] [B1] [M1] [A0]
		(ii)	Energy released = (binding energy of helium-3) - (binding energy of deuteriu $= 2.58 \times 3 - 1.11 \times 2$ [M1] $= 5.52$ MeV [A1]	
	(c)	**(i)**	 A1 for both elements on the curve	A1]
		(ii)	Helium-3 has a <u>higher binding energy per nucleon</u> than deuterium, so helium-3 has a <u>higher binding energy/ product is more stable/ higher mass defect</u>. Hence energy is released in the reaction.	[B1] [B1]

7	**(a)**	**(i)**	For the uniform wire, resistance of a segment of wire is proportional to the length of the segment. Hence by potential divider principle, $V_{ZY} = \frac{R_{ZY}}{R_{XY}} V_{XY} = \frac{L_{ZY}}{L_{XY}} V_{XY} = \frac{(6.0 - 2.0)}{6.0}(12.0)$ [C1] $= 8.0$ V [C1] Or $V_{XZ} = \frac{R_{XZ}}{R_{XY}} V_{XY} = \frac{L_{XZ}}{L_{XY}} V_{XY} = \frac{(2.0)}{6.0}(12.0)$ [C1] $= 4.0$ V [C1] Since point Y is earthed and hence at zero potential, potential at point Z = – 8.0 V [A1]
		(ii)	$R = \frac{\rho L}{A}$ where A is the cross-sectional area of the wire [M1] $A = \frac{\text{volume of wire}}{L}$ where volume of wire = $\frac{m}{D}$ [M1] Therefore $R = \frac{\rho L}{\left(\frac{m}{DL}\right)} = \frac{\rho D L^2}{m}$ (shown)
		(iii)	By $R = \frac{\rho D L^2}{m}$, since the length *L* is doubled while keeping the same cross-sectional area and material, the mass *m* will double. [M1] Or By $R = \frac{\rho L}{\text{A}}$, *L* is doubled while resistivity ρ and area A remains unchanged, [M1] Hence the <u>new resistance increases by a factor of 2</u> and the student is wrong. [A1]
	(b)	**(i)**	Increasing the p.d. across the thermistor results in <u>an increase in current, which in turn causes the temperature to rise</u>. [B1] The temperature rise causes <u>an increase in the number of (mobile) charge carriers</u>. [B1] The <u>effect of increase in number density of charge carriers outweighs the effect of increase in collisions between the free electrons and the lattice ions</u>. [B1] Hence, <u>resistance decreases</u> (i.e. the ratio of $\frac{V}{I}$ decreases). [B1]

		(ii)	 B1 for correct parallel circuit (diode direction is correct) B1 for correct circuit symbols (of LED, thermistor, cell and resistor) When temperature increases, the <u>resistance of the thermistor decreases</u> hence <u>potential difference across the thermistor will decrease</u> [M1] or When temperature increases, the resistance of the thermistor decreases hence the <u>effective resistance of the circuit decreases, by potential divider principle</u>, [M1] This causes <u>the potential difference across the LED to increase</u> beyond its threshold voltage hence turning it on. [A1]
		(iii)	The resistor is to prevent the LED from lighting up at low temperature. [B1]
	(c)	**(i)**	 M1 – correct shape A1 – correct label of 5.0 V
		(ii)	$V = V_0\left(1 - e^{-\frac{t}{RC}}\right)$ $2.0 = 5.0\left(1 - e^{-\frac{t}{3.0\times10^{6}(2.0\times10^{-6})}}\right)$ [M1] $t = 3.1\ \text{s}$ [A1]

8	**(a)**	A field of force is a region of space in which bodies may experience a force associated with the field. Or: A field of force is a region of space in which an object experiences a force. [A1]	
	(b)	**(i)**	Each ion will experience two forces that are equal in magnitude but opposite in direction, electric force (downwards) and magnetic force (upwards). [B1] Hence, the resultant force acting on the ion is zero that results in a straight-line motion. [B1]
		(ii)	The plate P is at a higher potential than plate Q. [A1] This forms a uniform electric field that is pointing downwards. Hence, the positive ions experience will experience a downwards electric force. [M1]
		(iii)	$F_E = F_B$ $qE = B_1 qv$ $v = \frac{E}{B_1} = \frac{3.0 \times 10^4}{0.25}$ [C1] $= 1.2 \times 10^5 \text{ m s}^{-1}$ [A1]
		(iv)	From the drawing, at the point of the entry into B_2, the ion experiences an upward resultant magnetic force that points towards the centre of the circle. [M1] By Fleming's left-hand rule, since the positive ion is moving to the right, the magnetic flux density would be pointing into the plane of paper. [A1]
		(v)	As the ion moves within B_2, it continues to experience a magnetic force that is perpendicular to the direction of its motion. [B1] Therefore, since work done is given by $W = Fs\cos\theta$, there is no work done by the magnetic force. [B1]
		(vi)	Within B_2 region, Magnetic force provides for centripetal force. [B1] $B_2 ev = m\frac{v^2}{r}$ [B1] $m = \frac{eB_2 r}{v}$ $m = \frac{eB_1 B_2 r}{E}$ (since $v = \frac{E}{B_1}$) [A0]

		(vii)	$m_x = \frac{qB_1B_2r_x}{E}$ $= \frac{1.6\times10^{-19}(0.25)(0.60)(0.074)}{3.0\times10^{4}}$ $= 5.92\times10^{-26}$ kg [A1]
		(viii)	$m_x = \frac{qB_1B_2r_x}{E}$ $\%\frac{\Delta m_x}{m_x} = \%\frac{\Delta B_1}{B_1} + \%\frac{\Delta B_2}{B_2} + \%\frac{\Delta r_x}{r_x} + \%\frac{\Delta E}{E}$ $= 1+1+(\frac{0.001}{0.074}\times100)+2$ [C1] $= 5.4\%$ [A1]
		(ix)	$\Delta r = 0.079 - 0.074 = 0.005$ m $\sigma = 0.001$ m [B1: correct Δr and σ] Since $\Delta r > 3\sigma$, the ions X and Y are distinguishable. [B1]
		(x)	Any of the following with proper explanation [B1, B1]. Decrease B_2. This increases radius separation for different masses, thus improving the resolution. Improve vacuum. This reduces collision of the ions with gas molecules. Use more precise detection surface. This reduces uncertainty in the radius.